

POSITIVE LOCAL COMPACTNESS AND UNIVERSAL HERMITE PROJECTIONS

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SOURCE AND TARGET

The source is Svela–Trapasso, *A quantum harmonic analysis approach to nonlinear time-frequency concentration*, arXiv:2605.28786 [1]. The authors formulate a broad open classification problem for smoothing and escape mechanisms of Cohen-class windows $S \in \mathcal{B}(L^2(\mathbb{R}^d))$. The relevant introductory passage is shown below.

compact perturbation of the identity

$$S = \text{Id} - \varphi_0 \otimes \varphi_0, \quad \varphi_0(y) = 2^{d/4} e^{-\pi|y|^2},$$

and also for every nontrivial time-frequency shift $S = \pi(z_0)$, $z_0 \neq 0$. Together with the previously established results for the Wigner and Born–Jordan distributions [67], this shows that the situation is significantly more challenging for noncompact operators.

While a full classification of smoothing and escape mechanisms in the Cohen class for $S \in \mathcal{B}(L^2(\mathbb{R}^d))$ remains an open problem (if at all possible), we contribute in this direction with an

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analysis of the structural properties behind the dichotomy. A key quantity in our analysis is the essential concentration value:

$$\Lambda_{p,\Omega}^{\text{ess}}(S) := \sup \left\{ \limsup_{n \rightarrow \infty} \|Q_S f_n\|_{L^p(\Omega)} : \|f_n\|_{L^2} = 1, f_n \rightharpoonup 0 \right\}.$$

Roughly speaking, this quantity measures the amount of concentration that can remain in Ω while the states escape weakly, and thus provides a sufficient criterion for both continuity and attainment. To be specific, in Proposition 5.2 we show that $\Lambda_{p,\Omega}^{\text{ess}}(S) < \Lambda_{p,\Omega}(S)$ is sufficient for an optimizer to exist, while in Proposition 5.3 we show that $\Lambda_{p,\Omega}^{\text{ess}}(S) = 0$ is equivalent to a weakly continuous concentration functional. In spite of the general picture, these results have interesting concrete applications: For instance, in Appendix A we use them to rule out exotic optimizers (cf. [67] for context) for the Wigner concentration problem over sufficiently small or large phase space balls.

A second structural contribution concerns the boundary between compactness and weak continuity: We prove that $\mathcal{B}_{\text{pt}}(p, \Omega) = \mathcal{K}$, where $\mathcal{B}_{\text{pt}}(p, \Omega)$ consists of windows for which $Q_S f_n(z) \rightarrow Q_S f(z)$ pointwise under weak convergence $f_n \rightharpoonup f$. We also show that $\mathcal{B}_{\text{wr}}(p, \Omega) \cap \mathcal{C}_c^\alpha = \mathcal{K}$.

The source proves several structural criteria. The first one used here is its positive localized-average criterion: if $S \geq 0$ and

$$H_{\Omega,S} := \chi_\Omega \star S = \int_\Omega \alpha_z(S) dz$$

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is compact, then $f \mapsto Q_S f$ is weak-to-norm continuous from $L^2(\mathbb{R}^d)$ into $L^p(\Omega)$, for every finite p . We first show that, for positive S , this local compactness condition is also necessary. The source also proves the endpoint formula $\Lambda_{\infty, \Omega}(S) = w(S)$, where $w(S)$ is the numerical radius, and numerical-radius attainment implies L^∞ -concentration attainment. Near the end of the same discussion, the source notes that sparse Hermite functions give noncompact positive diagonal examples for a fixed bounded observation set. The result below strengthens that observation to one Hermite projection in each dimension working simultaneously for every finite-measure observation window.

PROOF INTUITION

For positive windows the source's sufficient local compactness mechanism is exact. Indeed, if $S \geq 0$ and $S \in \mathcal{B}_{\text{wc}}(p, \Omega)$, then every normalized weakly-null sequence f_n satisfies $\|Q_S f_n\|_{L^p(\Omega)} \rightarrow 0$. Since $Q_S f_n \geq 0$, this forces

$$\langle H_{\Omega, S} f_n, f_n \rangle = \int_{\Omega} Q_S f_n \rightarrow 0.$$

A positive Hilbert-space operator with this vanishing property on all weakly-null sequences must be compact; otherwise an infinite-dimensional spectral subspace above a positive level supplies an orthonormal weakly-null countersequence.

The Hermite example then becomes an explicit noncompact positive window with compact localized average for every finite-measure Ω . The source's fixed-window Hermite argument only needs local decay of the ambiguity functions on the difference set $\Omega - \Omega$. In dimension d , take

$$u_n^{(d)} = \phi_n \otimes \phi_0 \otimes \cdots \otimes \phi_0.$$

The ambiguity function factors. Its first factor tends to zero away from $0 \in \mathbb{R}^2$, so the product tends to zero except on the codimension-two set $\{z_1 = 0\} \subset \mathbb{R}^{2d}$, which has measure zero. Dominated convergence gives local L^2 -decay on every ball. Diagonalizing over all balls then makes H_{Ω, S_d} compact for every bounded Ω .

If Ω has finite measure but is unbounded, truncate it by $\Omega_R = \Omega \cap B_R$. The bounded pieces give compact operators H_{Ω_R, S_d} , while

$$\|H_{\Omega, S_d} - H_{\Omega_R, S_d}\| \leq |\Omega \setminus B_R| \|S_d\| \rightarrow 0.$$

Thus H_{Ω, S_d} is a norm limit of compact operators. The source's localized-average criterion then gives weak-to-norm continuity and hence finite- p attainment by the direct method. At $p = \infty$, the same projection attains because every nonzero orthogonal projection attains its numerical radius.

DEFINITIONS

For $f \in L^2(\mathbb{R}^d)$ and $S \in \mathcal{B}(L^2(\mathbb{R}^d))$, write

$$Q_S f(z) = \langle \alpha_z(S) f, f \rangle, \quad z \in \mathbb{R}^{2d},$$

for the Cohen-class distribution in the quantum harmonic analysis formulation of [1]. For measurable $0 < |\Omega| < \infty$, the class $\mathcal{B}_{\text{wc}}(p, \Omega)$ consists of those windows for which

$$f_n \rightharpoonup f \implies \|Q_S f_n\|_{L^p(\Omega)} \rightarrow \|Q_S f\|_{L^p(\Omega)}.$$

The source proves that, for finite p , this agrees with weak-to-norm continuity of $f \mapsto Q_S f$ into $L^p(\Omega)$.

THEOREM A: POSITIVE WINDOWS

Let $S \in \mathcal{B}(L^2(\mathbb{R}^d))$ be positive, let $\Omega \subset \mathbb{R}^{2d}$ be measurable with $0 < |\Omega| < \infty$, and let $1 \leq p < \infty$. Then

$$S \in \mathcal{B}_{\text{wc}}(p, \Omega) \iff H_{\Omega, S} := \int_{\Omega} \alpha_z(S) dz \in \mathcal{K}.$$

Equivalently, for positive windows, weak-to-norm continuity over Ω is exactly compactness of the mixed-state localized average.

PROOF OF THEOREM A

The implication $H_{\Omega, S} \in \mathcal{K} \Rightarrow S \in \mathcal{B}_{\text{wc}}(p, \Omega)$ is the source paper's positive localized-average criterion.

Conversely, assume $S \in \mathcal{B}_{\text{wc}}(p, \Omega)$, and let $f_n \rightharpoonup 0$ be a normalized sequence in $L^2(\mathbb{R}^d)$. Since $S \geq 0$, also $\alpha_z(S) \geq 0$ and hence $Q_S f_n(z) \geq 0$. By weak-to-norm continuity,

$$\|Q_S f_n\|_{L^p(\Omega)} \rightarrow 0.$$

Because $|\Omega| < \infty$, Hölder's inequality gives $\|Q_S f_n\|_{L^1(\Omega)} \rightarrow 0$. Therefore

$$\langle H_{\Omega, S} f_n, f_n \rangle = \int_{\Omega} \langle \alpha_z(S) f_n, f_n \rangle dz = \int_{\Omega} Q_S f_n(z) dz \rightarrow 0$$

for every normalized weakly-null sequence f_n .

The operator $H_{\Omega, S}$ is positive. We claim that a positive bounded operator H on a Hilbert space is compact whenever $\langle H f_n, f_n \rangle \rightarrow 0$ for every normalized weakly-null sequence. If H were not compact, then by the spectral theorem there would be an $\varepsilon > 0$ such that the spectral projection $1_{[\varepsilon, \infty)}(H)$ has infinite-dimensional range. Choose an orthonormal sequence e_n in that range. Then $e_n \rightharpoonup 0$, while

$$\langle H e_n, e_n \rangle \geq \varepsilon$$

for all n , a contradiction. Thus $H_{\Omega, S}$ is compact.

THEOREM B: UNIVERSAL HERMITE PROJECTIONS

For every integer $d \geq 1$, there exists a noncompact positive orthogonal projection

$$S_d = \sum_{k=1}^{\infty} u_k^{(d)} \otimes u_k^{(d)} \quad \text{on } L^2(\mathbb{R}^d)$$

with the following property: for every measurable $\Omega \subset \mathbb{R}^{2d}$ with $0 < |\Omega| < \infty$ and every $1 \leq p < \infty$,

$$f_n \rightharpoonup f \text{ in } L^2(\mathbb{R}^d) \implies \|Q_{S_d} f_n - Q_{S_d} f\|_{L^p(\Omega)} \rightarrow 0.$$

Consequently $S_d \in \mathcal{B}_{\text{wc}}(p, \Omega)$, and the concentration supremum

$$\Lambda_{p,\Omega}(S_d) = \sup_{\|f\|_2=1} \|Q_{S_d} f\|_{L^p(\Omega)}$$

is attained for every finite-measure observation set Ω . In addition,

$$\Lambda_{\infty,\Omega}(S_d) = \sup_{\|f\|_2=1} \|Q_{S_d} f\|_{L^\infty(\Omega)}$$

is attained for every measurable $\Omega \subset \mathbb{R}^{2d}$ with positive measure. Thus these projections attain the concentration problem for every $1 \leq p \leq \infty$, while the weak-to-norm continuity conclusion is only claimed for finite p .

PROOF OF THEOREM B

Let $(\phi_n)_{n \geq 0}$ be the normalized one-dimensional Hermite basis. For $d \geq 1$, define

$$v_n^{(d)} = \phi_n \otimes \underbrace{\phi_0 \otimes \cdots \otimes \phi_0}_{d-1 \text{ copies}} \in L^2(\mathbb{R}^d).$$

These vectors are orthonormal as n varies. Write $z = (z_1, \dots, z_d) \in (\mathbb{R}^2)^d = \mathbb{R}^{2d}$, where each $z_j \in \mathbb{R}^2$. The tensor-product structure of the Schrödinger representation gives

$$|A(v_n^{(d)})(z)| = |A(\phi_n)(z_1)| \prod_{j=2}^d |A(\phi_0)(z_j)|.$$

In dimension one, the source records

$$A(\phi_n)(\zeta) = e^{-\pi|\zeta|^2/2} L_n(\pi|\zeta|^2),$$

and the consequences

$$A(\phi_n)(\zeta) \rightarrow 0 \quad (\zeta \neq 0), \quad |A(\phi_n)(\zeta)| \leq 1.$$

Therefore $A(v_n^{(d)})(z) \rightarrow 0$ for every z with $z_1 \neq 0$. The exceptional set $\{z \in \mathbb{R}^{2d} : z_1 = 0\}$ has Lebesgue measure zero. Since $|A(v_n^{(d)})(z)| \leq 1$, dominated convergence yields, for every integer $m \geq 1$,

$$a_{n,m}^{(d)} := \left(\int_{B_m^{2d}} |A(v_n^{(d)})(z)|^2 dz \right)^{1/2} \longrightarrow 0 \quad (n \rightarrow \infty),$$

where $B_m^{2d} \subset \mathbb{R}^{2d}$ is the centered radius- m ball.

Choose a strictly increasing subsequence n_k recursively so that

$$a_{n_k, m}^{(d)} \leq 2^{-k-m} \quad \text{for every } 1 \leq m \leq k.$$

This is possible because only finitely many conditions are imposed at stage k . Set $u_k^{(d)} = v_{n_k}^{(d)}$, $P_k = u_k^{(d)} \otimes u_k^{(d)}$, and define the strong operator sum

$$S_d = \sum_{k=1}^{\infty} P_k.$$

Since the $u_k^{(d)}$'s are orthonormal, S_d is the orthogonal projection onto their closed span. It is positive, $\|S_d\| = 1$, and it is not compact because $S_d u_k^{(d)} = u_k^{(d)}$ while $u_k^{(d)} \rightharpoonup 0$.

First let $E \subset \mathbb{R}^{2d}$ be bounded with $0 < |E| < \infty$. Choose m such that $E - E \subset B_m^{2d}$, and define

$$C_E(u)^2 = \int_E \int_E |A(u)(w - z)|^2 dz dw.$$

For $k \geq m$, the diagonal choice gives

$$\begin{aligned} C_E(u_k^{(d)})^2 &= \int_{\mathbb{R}^{2d}} |A(u_k^{(d)})(v)|^2 |E \cap (E - v)| dv \\ &\leq |E| \int_{E-E} |A(u_k^{(d)})(v)|^2 dv \leq |E| \int_{B_m^{2d}} |A(u_k^{(d)})(v)|^2 dv \\ &\leq |E| 2^{-2k-2m}. \end{aligned}$$

Therefore $\sum_k C_E(u_k^{(d)}) < \infty$.

Let $H_{E,k} = \chi_E \star P_k = \int_E \alpha_z(P_k) dz$. The source computes

$$\|H_{E,k}\|_{S^2} = C_E(u_k^{(d)}),$$

so $\|H_{E,k}\|_{\mathcal{B}} \leq C_E(u_k^{(d)})$. Since $H_{E,k}$ is compact for every k , the series

$$H_{E,S_d} = \int_E \alpha_z(S_d) dz = \sum_{k=1}^{\infty} H_{E,k}$$

converges in operator norm to a compact operator. Thus $H_{E,S_d} \in \mathcal{K}$ for every bounded finite-measure E .

Now let $\Omega \subset \mathbb{R}^{2d}$ be arbitrary with $0 < |\Omega| < \infty$. Put $\Omega_R = \Omega \cap B_R^{2d}$. Each H_{Ω_R, S_d} is compact by the bounded case. Moreover, for every measurable $F \subset \mathbb{R}^{2d}$ of finite measure,

$$\|H_{F, S_d}\| = \left\| \int_F \alpha_z(S_d) dz \right\| \leq \int_F \|\alpha_z(S_d)\| dz = |F| \|S_d\|.$$

Consequently

$$\|H_{\Omega, S_d} - H_{\Omega_R, S_d}\| = \|H_{\Omega \setminus B_R^{2d}, S_d}\| \leq |\Omega \setminus B_R^{2d}| \|S_d\| \longrightarrow 0.$$

The operator H_{Ω, S_d} is therefore a norm limit of compact operators, hence compact.

By Theorem A, or equivalently by the source’s positive localized-average criterion applied to the compact operator H_{Ω, S_d} , this positive S_d satisfies, for every $1 \leq p < \infty$,

$$f_n \rightharpoonup f \implies \|Q_{S_d} f_n - Q_{S_d} f\|_{L^p(\Omega)} \rightarrow 0.$$

This proves $S_d \in \mathcal{B}_{wc}(p, \Omega)$ for every finite-measure Ω .

For finite p , weak-to-norm continuity gives attainment by the direct method. The closed unit ball of $L^2(\mathbb{R}^d)$ is weakly compact, $f \mapsto \|Q_{S_d} f\|_{L^p(\Omega)}$ is weakly continuous on it, and $S_d \neq 0$ makes the maximum positive. Since Q_{S_d} is homogeneous of degree two, every nonzero maximizer over the closed unit ball can be normalized to the unit sphere. Thus $\Lambda_{p, \Omega}(S_d)$ is attained for $1 \leq p < \infty$.

It remains only to record the endpoint $p = \infty$. The source proves that $\Lambda_{\infty, \Omega}(S) = w(S)$ for every positive-measure Ω , and that numerical-radius attainment of S implies endpoint concentration attainment. Since S_d is an orthogonal projection, $w(S_d) = \|S_d\| = 1$, and the numerical radius is attained at every unit vector in $\text{Ran } S_d$. Hence $\Lambda_{\infty, \Omega}(S_d) = 1$ is attained for every positive-measure Ω .

VERIFICATION AND NOVELTY CHECK

The proof is noncomputational. It uses the source paper’s positive localized-average criterion, its L^∞ numerical-radius formula, the Hermite ambiguity decay quoted in the source, and the elementary tensor factorization of ambiguity functions. The new structural step is the converse in Theorem A: for positive windows, compactness of $H_{\Omega, S}$ is necessary as well as sufficient for finite- p weak continuity. The new Hermite steps are the all-dimensional tensor-Hermite local decay observation, the simultaneous diagonal choice over all balls, and the finite-measure truncation argument for H_{Ω, S_d} .

Cheap run indexes were searched for arXiv:2605.28786, the exact title, ‘positive localized average’, ‘compact localized average’, ‘Hermite projection’, ‘positive diagonal’, ‘finite measure’, ‘all dimensions’, and related phrases. No existing packet or attempt for this strengthened result was found. A bounded web search for the exact source title, positive localized-average converse phrases, and strengthened Hermite finite-measure phrases found the arXiv source record and no later answer.

LIMITATIONS

This is not a full classification of all Cohen-class windows. It does give an exact finite- p weak-continuity classification for positive bounded windows, and one explicit noncompact positive projection in each dimension that works for every finite-measure observation window in that dimension. It does not classify arbitrary nonpositive windows. The endpoint $p = \infty$ is

covered for attainment of the Hermite projections only, not by weak-to-norm continuity.

HUMAN REVIEW RECOMMENDATION

First review Theorem A, especially the spectral-theorem argument that positivity plus vanishing quadratic forms on normalized weakly-null sequences forces compactness. Then review the tensor factorization of the ambiguity functions and the dominated-convergence step on balls in $\mathbb{R}^{2d}$. Then review the estimate of $C_E(u_k^{(d)})$ for bounded E , the norm convergence of $H_{E,S_d} = \sum_k H_{E,k}$, and the truncation step

$$H_{\Omega,S_d} = \lim_{R \rightarrow \infty} H_{\Omega \cap B_R^{2d}, S_d}$$

for unbounded finite-measure Ω . If these are accepted, the finite- p result follows directly from the source paper's positive localized-average criterion. For the endpoint, check that S_d attains its numerical radius as an orthogonal projection, and apply the source paper's L^∞ numerical-radius formula.

REFERENCES

- [1] E. A. T. Svela and S. I. Trapasso, *A quantum harmonic analysis approach to nonlinear time-frequency concentration*, arXiv:2605.28786, 2026.